



Radiographer research engagement: An international mixed methods study

A. Stewart-Lord ^{a,*}, K. Matthews ^{b,c}, A. Partner ^d, B. Mussmann ^{e,f,g}

^a College of Health and Life Sciences, School of Allied Health and Life Sciences, London South Bank University, 103 Borough Road, London, SE1 0AA, England

^b Monash Centre for Scholarship in Health Education, Faculty of Medicine, Nursing and Health Sciences, 27 Rainforest Walk, Monash University, Clayton, Victoria, 3800, Australia

^c Department of Radiation Oncology, Peter MacCallum Cancer Centre, 305 Grattan Street, Melbourne, Victoria, 3800, Australia

^d School of Health Sciences, University of Nottingham, Medical School Building, Queens Medical Centre Campus, Nottingham, NG7 2HA, United Kingdom

^e Department of Radiology, Odense University Hospital, Sdr. Boulevard 29, 5000 Odense C, Denmark

^f Department of Clinical Research, University of Southern Denmark, Campusvej 55, 5220, Odense, Denmark

^g Department of Life Sciences and Health, Oslo Metropolitan University, Pilestødet 48, N-0130 Oslo, Norway

ARTICLE INFO

Article history:

Received 29 October 2025

Received in revised form

27 January 2026

Accepted 9 February 2026

Available online 12 March 2026

Keywords:

Research culture

Professional issues

Radiography

ABSTRACT

Introduction: The development of skilled and motivated research-active radiographers is a means to improve patient care and service delivery. For this to be realised, embedding a culture of enquiry, critical thinking, and research-based activities by radiographers is essential. This study aimed to provide an analysis of the Radiography research culture internationally and identify the current barriers, facilitators and potential future solutions to promote radiography research career pathways.

Methods: A sequential mixed methods design was utilised with participants from across the international radiography workforce. Data collection occurred during two phases. Phase 1 - Quantitative cross-sectional survey (n=182) and Phase 2 - Semi-structured interviews (n=14). Descriptive and inferential statistics were used to analyse quantitative data while qualitative data were analysed using reflective thematic analysis approach.

Results: Strong barriers to research participation included lack of protected time, funding, and management support, while key enablers were job satisfaction, a solvable problem, departmental research activity, and mentorship.

Seven themes were created from the qualitative data: 1) Developing research skills; 2) Evolution of the role, a different path; 3) From participating to leading, a research journey; 4) Money talks; 5) Outputs vs effort; 6) Research interest and professional engagement; 7) Clinical vs academic roles.

Conclusion: The international radiographer community desires research career pathways and opportunities to contribute to research, however significant cultural and organisational factors continue to limit research involvement. Research careers are often self-directed with limited guidance or support. International radiographer research networks may also enable research career trajectories.

Implications for practice: Radiographer engagement in research is difficult to realise internationally. Organisational and cultural investment is essential to enable radiographer research. Support and mentorship are essential to research growth.

© 2026 The College of Radiographers. Published by Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

* Corresponding author.

E-mail addresses: stewara2@lsbu.ac.uk (A. Stewart-Lord), kristie.matthews@monash.edu (K. Matthews), alexandra.partner@nottingham.ac.uk (A. Partner), bo.mussmann@rsyd.dk (B. Mussmann).

Introduction

In June 2021, the College of Radiographers (CoR) published its new research strategy,¹ aiming to raise the impact and profile of radiography through high-quality research focused on improving patient care and service delivery and expanding radiography research capacity through the development of skilled and motivated research-active professionals. As part of a wider group of Allied Health Professionals (AHPs), radiographers can drive sustainable change through research activities that impact processes, pathways, technology, and people. This requires promoting a culture of enquiry and critical thinking that embraces change whilst encouraging, supporting, and developing research-based activities within clinical departments. Similarly, one of the important aims of the European Federation of Radiographer Societies (EFRS) is to promote evidence-based practice in radiographic practice. In the 2023-25 activity plan², EFRS included strategic efforts to promote and facilitate radiographer-led academic and non-academic research. In order to achieve this goal, it is imperative to first establish the current level of radiographer research engagement.

Literature Review

Barriers to research involvement include a combination of factors, rather than one single reason.³ A recent systematic review⁴ explored the barriers to evidence-based practice and research implementation in clinical Radiography practice internationally. This mixed-method systematic review included 376 studies dating from 2003 to July 2023. Results showed that there was a high level of motivation to engage in research globally, however, numerous barriers were reported such as insufficient time allocation for research, lack of resources, lack of research culture and inadequate research-related skills and knowledge. The review concluded with the need to promote a research culture within Radiography as a profession. It recognised the importance of an environment valuing interdisciplinary collaboration and innovation, with a management structure to enhance research development training. The impact of a positive research culture on implementation of research outcomes was highlighted by Bolejko et al⁵ and Ooi et al⁶ Furthermore, a narrative review⁷ found that radiographer-led research contributed to decreased radiation exposure to patients and staff, while maintaining image quality.

Whereas the number of radiographers involved in research has indeed increased,⁸ leading to an increased number of peer-reviewed publications led by radiographers over the last decade, a Danish review highlighted that the majority of publications originate from a relatively small group of individual radiographers.⁹ The finding was also reflected in numerous world-wide publications stating positive attitudes toward research, but little actual involvement¹⁰⁻¹³. Some stress that limited experience, knowledge and self-confidence hinder involvement¹⁴⁻¹⁷ and one African study mentioned language barriers, publication costs and brain drain from low and middle-income countries as major challenges to research involvement¹⁸. Many barriers can be counteracted by dedicated research radiographers who may be able to facilitate features such as staff development^{16,17}.

With the imperatives from the CoR and EFRS to grow radiography research, this study aimed to provide a robust analysis of the Radiography research culture internationally and identify the current barriers, facilitators and potential future solutions to promote radiography research career pathways.

Methods

A sequential mixed methods design¹⁹ was utilised to gain an understanding of the landscape of research amongst the radiography profession internationally, and to identify the barriers and facilitators in research engagement across the professional radiography workforce. Data collection was conducted across two phases. Integration in a mixed methods study can take place at three levels (design, methods and interpretation).²² In this study integration took place at all three levels: first at design level to address the specific objective of identifying barriers and facilitators to research; second at methods level where the key findings from the survey informed the development of the questions asked at interview; and finally at interpretation level where the discussion incorporates both findings from the qualitative and quantitative phases. This explanatory approach allowed the qualitative phase to build on the initial quantitative phase.²³

Phase 1 - Quantitative cross-sectional survey

The questionnaire included 20 items focusing on Research and Development Culture Index, barriers and facilitators to participating and leading research, and factors enabling participating in or leading research. The questionnaire was modified (terminology only) from an existing validated questionnaire used in the National Institute for Health Research (NIHR) incubator project for diagnostic and therapeutic radiographers in the UK.¹² Content validity was tested by review from an expert panel of academics and radiographers. A pilot study was done to ensure the questionnaire was measuring the appropriate constructs (face validity). The questionnaire was delivered in an online format through onlinesurveys.ac.uk.

Phase 2 - Semi-structured interviews

The aim of this phase was to utilise qualitative enquiry to explore radiographer research career pathways. Results from phase one were used to formulate a semi structured interview guide in consultation with our expert panel. Two pilot interviews were conducted with members of the International Advisory Board for the journal Radiography to ensure trustworthiness and rigour in our approach. No changes were made post pilot.

Ethical considerations

Ethical approval was gained through a UK University Ethics Panel. All participants were required to complete a consent form. Survey responses were anonymous, and interview data were anonymised and kept securely on password protected cloud-based servers.

Recruitment and sample

Phase 1

The survey was distributed through the professional bodies for Radiography the Society and College of Radiographers (SCoR) and the European Federation of Radiographer Societies (EFRS) membership. The questionnaire was open from June – August 2024. Professional bodies utilised their own distribution approach. The link to the online survey was sent to members through membership networks and promotional materials. We cannot confirm how

many members received the invitation. Reminder emails were sent to network links twice during the data collection period in months 1 and 2. Social media was also used by the research team and professional bodies to promote the study

Phase 2

Recruitment to phase 2 was initiated in the phase 1 questionnaire where participants were asked if they were willing to participate in the phase 2 interviews. Respondents who were willing to participate were purposively selected from different countries and practice roles. Interviews were carried out online using software (i.e. MS Teams) from October – December 2024. Interviews were recorded and professionally transcribed and then de-identified.

Data analysis

Phase 1

Statistical analyses were performed by author BM using Stata 18/BE (StataCorp, College Station, TX, USA). A two-tailed significance level of $p < 0.05$ was applied throughout. To examine the association between various aspects of research culture and the likelihood of involvement in research activities, multivariate logistic regression was conducted. The dependent variable was research involvement, dichotomized as 1 for respondents who reported having participated in at least one research activity, and 0 for those who had not. The independent variables included a range of categories reflecting the research culture, such as access to support, opportunity for development, confidence, and attitudes toward learning about research.

To assess model fit, the Hosmer–Lemeshow goodness-of-fit test was used.²⁰ Additionally, the model's explanatory power was evaluated using Nagelkerke's pseudo R-squared, ranging from 0 to 1, with higher values indicating greater explanatory strength.

Phase 2

The data were analysed using Braun and Clarke's reflective thematic analysis approach.²¹ There are six phases to the process, which are used in an iterative way, rather than purely linear: 1. Dataset familiarization; 2. Data coding; 3. Initial theme generation; 4. Theme development and review; 5. Theme refining, defining, and naming; 6. Writing up. ASL and BM initially undertook stages 1, 2 and 3. ASL, KM and AP then iteratively collaborated to complete the remaining stages. The research team background collectively represented four countries, two professions, and clinical and academic settings, hence team reflexivity was sustained by regular conversations during data analyses to examine individual interpretations. NVivo (Version 14) was used to facilitate the analysis process.

Results

Quantitative findings

In the survey, 188 replies from 45 countries and all world regions were recorded. Six replies were removed from the dataset because respondents were not radiographers (3 medical physicists and 3 MD). Consequently, 182 were included in the analysis. Most responses came from diagnostic radiographers (Table 1) in European countries (n=134), mainly from the United Kingdom with 31 % of the responses (Table 2). In many countries, only 1-3 responses were obtained. Furthermore, only two responses were recorded from the Americas, with a similar number from Oceania.

Table 1
Radiographic specialties reported. No. of observations=193.

Specialty	n
Diagnostic radiography	84
Nuclear medicine	14
Radiation therapy	29
Did not report	66
Two or more specialties	9

Table 2
Country of origin for respondents. 'Other' had 3 or less respondents (34 countries). A full list of countries can be found in Appendix A.

Country	n	%
United Kingdom	56	31
Portugal	11	6
Singapore	9	5
Malta	8	4
Denmark	7	4
Ireland	6	3
Italy	6	3
Nigeria	6	3
Norway	6	3
Greece	5	3
South Africa	4	2
Spain	4	2
Other	54	28
Total	182	100

The percentage of responses/non-responses could not be calculated because of the method of distribution. The respondents were predominantly female (n=113 / 62 %). Mean age was 43.6 years (SD 10.0) ranging from 23 to 65 years. Respondents reported a mean experience of 17.9 years (SD 10.4) ranging from half a year to 43 years. Aside from clinical examinations, the responses covered multiple professional roles such as advanced practitioner (n=26), researcher (n=48), service manager (n=7), lecturer or senior lecturer (n=50)

The replies to each question, including frequencies and percentages, are listed in Appendix B. The multivariate logistic regression examining the influence of research culture factors on research involvement identified two statistically significant predictors:

- Having the opportunity to develop professional practice (OR = 3.14, $p = 0.030$), and
- Being open to learning about research (OR = 2.28, $p = 0.025$).

The model demonstrated moderate explanatory power, with a Nagelkerke's pseudo R^2 of 0.27, suggesting that approximately 27 % of the variance in research involvement was explained by the model. Furthermore, the Hosmer–Lemeshow goodness-of-fit test ($p = 0.3876$) indicated that the model fits the data well. The full list of odds ratios for research cultural factors' influence on being involved in research can be viewed in Appendix C.

Lack of protected time at work and other work roles taking priority were the item that most radiographers agreed or strongly agreed being barriers to participating in research with 88.8 % and 87.7 %, respectively, followed by lack of funding (81.5 %), lack of management support (67.4 %), Lack of research supervision (63.3 %), and lack of own research skills (48.4 %). The most important factors enabling participation in research were *increased job satisfaction* and *having identified a problem that needs solving*. Furthermore, research activity in the department and

Table 3
Sample of participants for semi-structured interviews.

Participants	Profession	Countries (number)
RC1 – 4	Diagnostic Radiographer / Imaging Clinical	Saudi Arabia (1) Sweden (1) UK (2)
RT1-3	Therapeutic Radiographers / Radiation therapist Clinical	Poland (1) Portugal (1) UK (2)
RU1-7	All radiographers university / academia	Ghana (1) Norway (1) South Africa (2) UK (3)

availability of research mentorship was considered important by the respondents (Appendix D).

Qualitative findings

A final sample size of n=14 interviews were conducted to achieve the appropriate balance of richness of data and variability in the sample. (See Table 3). Participants represented 8 countries across clinical and academic settings. Data demonstrated breadth in prior research experience from participants, although it is acknowledged those who volunteered for interviews all appeared to be invested in growing research capacity for their profession.

Analysis of the qualitative data identified seven interrelated themes that explained the features, barriers and enablers of a

radiographer research career pathway. The seven themes were *Developing research skills; Evolution of the role, a different path; From participating to leading, a research journey; Money talks; Outputs vs effort; Research interest and professional engagement; and Clinical vs academic roles*. Each theme is described in Table 4

Discussion

The quantitative and qualitative methods produced valuable insights into the current culture of radiographer research from an international perspective and are integrated²² within the discussion. Overall conclusions are drawn from the meta-inferences (the merging of quantitative and qualitative inferences) resulting in concordance,²² however, there was no intent to explore relational, predictive or causal relationships.²⁷ Globally, radiographers see research engagement as worthwhile and a professional expectation, and yet similar barriers exist across jurisdictions. Despite research being high on the agenda for both the College of Radiographers and EFRS there are particular challenges with access to education, mentorship and training opportunities.

The results indicate that respondents in the current study were substantially more involved in research activities than those in earlier reports, likely attributable to self-selection bias. Specifically, 61 % reported having participated in at least one research activity, and 60 % held an MSc or PhD degree. This contrasts with findings from a Nordic study, where only 31 % of radiographers had been involved in research—primarily in data collection roles that were not initiated by radiographers themselves.¹⁰ Similarly, a recent

Table 4
Themes from qualitative data.

Theme	Description	Illustrative data
Developing research skills	Radiographers need to self-direct their engagement in formal training, mentoring, educational pathways, and informal skills development in order to become research active. Access to training can be difficult.	"Your PhD – gives you almost a licence to drive research. It feels like it's a kind of tick box, but a very important tick box but it means that you can then step up to that next stage." (RC1)
Evolution of the role, a different path	Research roles are not common, and radiographers seeking a research career pathway need to identify their own opportunities. Role models and organisational support are enabling to taking a different path.	"I think without meeting the people I met, I would not have been in the position I am right now... my main supervisor, he was instrumental in guiding me and encouraging me because we are maybe like four or five radiography [PhD] students in the whole of the country" (RC2)
From participating to leading, a research journey	Radiographers build their confidence to lead research over time. Feeling an imposter remains a challenge where research is dominated by others.	"I still kind of can't believe I'm here sometimes. I mean imposter syndrome is massive ... I don't think our profession is as well-known [for research] as others, and that that can be an issue sometimes" (RU2)
Money talks	Funding for radiographers to do research in the form of grants and within employment frameworks are essential. However, funding opportunities and organisational support are inconsistent.	"There needs to be a commitment from a high level [of the clinical service] to enable research to be involved at every level. But there isn't at the moment: it's just seen as a luxury ... So there needs to be a change, but I think it needs to be at the high level" (RT3)
Outputs vs effort	Outputs are valued by radiographer researchers to disseminate evidence-based practice to the profession. However, there are limited allowances for the effort required to do so.	"I think it's just trying to dedicate the time to do it ... not just undertaking the research, but disseminating the information and not just thinking that the project's finished once the data's collected. So, I think I probably found that quite challenging" (RU6)
Research interest and professional engagement	Radiographer research interest and engagement is influenced by the degree research is valued by the profession. Professional body advocacy of research engagement is seen as key to growth.	"People do still say, 'What's in it for me?' ... And I will say to them, 'No, you're doing it for your profession. You want your profession to grow and you want the status of your profession to grow, and, if you do that with your research, then you will grow'" (RU5)
Clinical vs academic roles	A well-defined research radiographer career pathway is challenging in both clinical and academic settings. Expectations of clinical or teaching activities are commonly in tension with research activities.	"I worked as a lecturer in one of the universities, my goal was to be in the academic field and work on research more and publish more articles and studies. But after one year I did not find myself doing research... for some reason it was really overwhelming, maybe because I had a lot of courses to teach" (RC4)

British study¹² found that 41 % of radiographers had participated in research to some extent. In both studies, the respondents' educational levels were generally lower than those in the current sample.

More radiographers are finding routes around the hurdles and accessing enablers to get involved in research. Regarding the influence of cultural factors, only 'openness to learning about research' and 'opportunities to develop professional practice' resulted in statistically significant Odds Ratios. Some other factors resulted in OR less than 1 which may seem counterintuitive in most cases because many respondents agreed to the importance of those factors' influence. However, none of those ORs were statistically significant and may be caused by low statistical power. Thus, they should be interpreted with great caution.

Facilitators and barriers identified in the current study are not directly comparable to those in earlier research. However, themes such as protected time and collegial support emerged as key facilitators, while barriers such as a lack of knowledge and skills aligned with other study findings.^{4,15} Barriers to involvement were also reported in a systematic review⁴ which looked at 31 studies and concluded that obligations, motivation and organisational culture, skills gaps, resources, opportunities and governance were all providing challenges. The authors call upon clinical radiography managers to tackle these by not only protecting time but also working on initiatives to address governance and culture.⁴ Moreover, the barriers and enablers identified in the UK by Knight et al.¹² were echoed in the broader international context of the current study.

Qualitative findings showed that research skill development remains linked to a formal education either through master's or doctoral studies correlating with findings from other studies.²⁴ Even though there are more radiographers with doctoral qualifications who could drive forward the research agenda²⁵ the lack of opportunities to train remains limited, with many participants having to train abroad to obtain a suitable research degree. Research skill development was often associated with being exposed to research through being part of a research team, however findings showed that accessing suitably experienced and qualified mentors who could provide guidance and support remains difficult. Radiographers reported needing someone to bounce ideas off to problem solve, as well as the emotional support of peers to move away from the traditional clinical role into research. There is a need for international networks to help facilitate collaborative research working and provide active peer support. This was particularly evident from participants in countries where established national networks did not exist and the opportunity to learn from those countries with more established networks were valued.

No development or educational pathway was without challenges, and, similar to the findings by Knight et al.,¹² employer support was the most notable barrier. Most research careers or developmental pathways were self-directed; however, although participants were able to share examples of mentorship or support gained from other professionals, there was also a need for this support to be more formalised. A lack of mentors and peer support meant that problems with confidence, motivation and self-doubt couldn't be addressed. Participants perceived they lacked confidence compared to other professional fields.²⁶ Even participants who were research active and confident in leading research continued to struggle with 'imposter syndrome', as noted in other studies.^{12,24} Research in other health professions²⁸ often focus on the prevalence of imposter syndrome and associated factors rather than identifying ways in which to resolve this. This study was able to identify specific ways in which individuals can be supported i.e. mentorship, peer support and international networks.

Funding - and the lack thereof - was noted by most participants as the most challenging barrier to overcome in their research engagement. It was noted that the access to funding streams have grown in recent years, however the lack of available time and skills to write successful funding grants remains challenging. Funding is also directly linked to having funded time to be released from clinical duties to engage in grant writing.²⁴ This is also true for those radiographers who are seeking time to produce research outputs such as writing for publication. Publishing research outputs can be difficult, particularly when publishing in English as a second language. Factoring in writing and publishing time to any research project is a key milestone and time consideration, and one that should be supported by line managers and organisations.

There is an assumption that all practice should be underpinned by evidence-based research, however very few radiographers are involved in producing evidence through research. Findings from this study not reported elsewhere showed that professional body advocacy of research engagement is seen as key to growth. There is a notion amongst participants that working across both academic and clinical sectors could afford radiographers the best chance of being research active. This option should be more readily available to radiographers, and participants urged the varied sectors to consider this. All participants valued research engagement for radiographers and were positive in promoting this career pathway, however they noted changes were required to support such a career choice.

This study has some limitations. The distribution method precluded follow-up of non-respondents and offered no control over the number of responses, leaving the response rate unknown. Convenience sampling and small, and uneven samples across countries may limit the representativeness of the findings and their generalisability to the global radiographer population. In particular, respondents from the UK were highly represented in both qualitative and quantitative data which may have been influenced by the recruitment strategy. Additionally, the results may be affected by self-selection bias, with participation by those radiographers who are interested in a research career.

Conclusions

The findings of this study provide an understanding of radiographer research engagement internationally. The two phases present a global picture of the current state of radiographer research engagement, as well as the facilitators and barriers to research. Data showed that openness to learning about research and opportunities to develop professional practice were enabling, however significant cultural and organisational factors continue to limit research involvement. Research careers are often self-directed with limited guidance or support, and there needs to be significant investment and advocacy from organisations and professional bodies if this paradigm is to shift. Growing international radiographer research networks to help facilitate collaborative research working may also enable research career trajectories.

Ethics approval and consent to participate

Ethical approval was gained through London South Bank University HSCSEP Ethics Panel ETH2324-0122 (Stage 1) and ETH2425-0051 (Stage 2). The project did not require Health Research Authority approval. All interview participants for Stages 1 and 2 were provided with participant information sheets and given time to decide. They all completed written consent forms prior to interview. All interview data were anonymised and kept securely on password protected cloud based servers.

Author contributions

ASL: Conceptualisation, Methodology, Formal Analysis, Investigation, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualisation, Project Administration

BM: Conceptualisation, Methodology, Formal Analysis, Investigation, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualisation, Project Administration

KM: Formal Analysis, Writing - Original Draft, Writing - Review & Editing

AP: Formal Analysis, Writing - Original Draft, Writing - Review & Editing

Availability of data

Data required for this study may be made available by the author(s) upon reasonable request.

Declaration of Generative AI and AI-assisted technologies in the writing process

NA.

Funding

This research was *part-funded* by the *Elsevier Radiography* as part of the Editorial Fellowship. We acknowledge the funding from Elsevier, for transcription services which enabled the qualitative data analysis.

Conflict of interest statement

Adele Stewart-Lord and Bo Mussman declare that they were employed as Editorial Fellow for Radiography for the duration of this research project. Adele Stewart-Lord and Kristy Matthews declare current Editorial roles in Radiography.

Acknowledgements

We gratefully acknowledge the opportunity and support through our Editorial Fellowship with the journal Radiography in addition to the Society and College of Radiographers (SCoR) and European Federation of Radiography Societies (EFRS).

We also acknowledge the invaluable input from members of the International Advisory Board, who gave their time and shared their expertise in the design and development of the project.

We are truly indebted to the participants who generously gave their time and shared their experiences during the questionnaire and interviews for the project.

Appendices

The replies to each question, including frequencies and percentages, are listed in [Appendix B](#). The multivariate logistic regression examining the influence of research culture factors on

Appendix A

Full list of participating countries

Country	Freq.	Percent
Albania	2	1.10
Algeria	2	1.10
Australia	1	0.55
Austria	3	1.65
Bahrain	1	0.55
Bangladesh	1	0.55
Belgium	1	0.55
Bosnia Herzegovina	1	0.55
Croatia	1	0.55
Cyprus	1	0.55
Denmark	7	3.85
Estonia	1	0.55
France	1	0.55
Ghana	2	1.10
Greece	5	2.75
India	1	0.55
Indonesia	1	0.55
Ireland	6	3.30
Israel	1	0.55
Italy	6	3.30
Latvia	2	1.10
Lithuania	3	1.65
Malaysia	1	0.55
Malta	8	4.40
Mexico	2	1.10
Myanmar	1	0.55
New Zealand	1	0.55
Nigeria	6	3.30
North Macedonia	1	0.55
Norway	6	3.30
Pakistan	1	0.55
Philippines	3	1.65
Polska	1	0.55
Portugal	11	6.04
Saudi Arabia	1	0.55
Singapore	9	4.95
Slovenia	3	1.65
South Africa	4	2.20
Spain	4	2.20
Sri Lanka	1	0.55
Sweden	1	0.55
Switzerland	2	1.10
The Netherlands	1	0.55
United Kingdom	56	30.77
Yemen	1	0.55
Not specified	7	3.85
Total	182	100.00

research involvement identified two statistically significant predictors:

- Having the opportunity to develop professional practice (OR = 3.14, $p = 0.030$), and
- Being open to learning about research (OR = 2.28, $p = 0.025$).

The model demonstrated moderate explanatory power, with a Nagelkerke's pseudo R^2 of 0.27, suggesting that

approximately 27 % of the variance in research involvement was explained by the model. Furthermore, the Hosmer–Lemeshow goodness-of-fit test ($p = 0.3876$) indicated that the model fits the data well. The full list of odds ratios for research cultural factors' influence on being involved in research can be viewed in [Appendix C](#).

Appendix B

Replies to questionnaire items regarding Research and Development Culture Index

Item	Strongly disagree, n (%)	Disagree, n (%)	Agree, n (%)	Strongly agree, n (%)
Development of my professional practice/role is valued as part of my job (n=177)	12 (6.78)	24 (13.56)	81 (45.76)	60 (33.90)
There are people around to help and support me to change and/or develop professional practice (n=178)	9 (5.06)	34 (19.10)	107 (60.11)	28 (15.73)
There is strong professional leadership (n=177)	19 (10.73)	49 (27.68)	79 (44.63)	30 (16.95)
There is opportunity to develop professional practice in my area (n=177)	9 (5.08)	40 (22.60)	91 (51.41)	37 (20.90)
There are regular staff meetings to explore ideas (n=176)	30 (17.05)	66 (37.50)	66 (37.50)	14 (7.95)
I have access to training and development opportunities which give me the skills to question and investigate practice (n=178)	18 (10.11)	33 (18.54)	94 (52.81)	33 (18.54)
There are opportunities to reflect on my work/practice (n=177)	17 (9.6)	40 (22.6)	95 (53.67)	25 (14.12)
The development work that I do links with the Directorate's (or division /institutions) plans (n=172)	11 (6.4)	41 (23.84)	94 (54.65)	26 (15.12)
I understand research terminology (n=176)	7 (3.98)	14 (7.95)	93 (52.84)	62 (35.23)
I feel confident about using research in professional practice (n=176)	11 (6.25)	23 (13.07)	86 (48.86)	56 (31.82)
I know how professional practice is influenced by research (n=176)	6 (3.41)	7 (3.98)	91 (51.7)	72 (40.91)
I would like to learn about research activity during the next 6 months (n=178)	6 (3.37)	18 (10.11)	79 (44.38)	75 (42.13)
I would like more opportunities to share professional practice (n=176)	2 (1.14)	10 (5.68)	90 (51.14)	74 (42.05)
Development ideas and/or research and/or information across my institution (n=172)	5 (2.91)	19 (11.05)	101 (58.72)	47 (27.33)
I am very keen to use research in professional practice (n=175)	3 (1.71)	7 (4.00)	87 (49.71)	78 (44.57)
My discipline here works as equal partners with other disciplines in order to change or develop practice (n=177)	16 (9.04)	46 (25.99)	83 (46.89)	32 (18.08)

Appendix C

Multivariate logistic regression model of research cultural factors' influence on being involved in research. Hosmer–Lemeshow Goodness-of-Fit test $p=0.376$; Nagelkerke's pseudo $R^2= 0.27$, no. of observations= 158. CI, confidence interval

Factor	Odds Ratio	p-value	95 % CI
Development of professional practice/role is valued	0.85	0.722	0.35 to 2.06
Help and support	1.51	0.416	0.56 to 4.09
Strong professional leadership	0.85	0.736	0.34 to 2.16
Opportunity to develop professional practice	3.14	0.030	1.12 to 8.85
Regular staff meetings to explore ideas	0.45	0.062	0.19 to 1.04
Access to training and development opportunities	1.53	0.355	0.62 to 3.77
Opportunities to reflect on work/practice	0.51	0.183	0.19 to 1.38
Development work links with Directorate's plans	0.78	0.604	0.30 to 2.02
Understanding research terminology	1.70	0.354	0.55 to 5.21
Confidence about using research	1.59	0.356	0.59 to 4.28
Knowing how professional practice is influenced by research	0.82	0.668	0.34 to 2.00
Being open to learn about research	2.28	0.025	1.11 to 4.71
Being open to opportunities to share professional practice	1.24	0.680	0.44 to 3.52
Developing ideas across institution	0.48	0.110	0.20 to 1.18
Being keen to use research in professional practice	0.94	0.892	0.40 to 2.22

Appendix D
Factors enabling participation in or leading research

Item	Strongly disagree, n (%)	Disagree, n (%)	Agree, n (%)	Strongly agree, n (%)
Increased job satisfaction (n=177)	3 (1.7)	10 (5.6)	87 (49.2)	77 (43.5)
A problem identified that needs changing (n=176)	2 (1.1)	3 (1.7)	95 (54)	76 (43.2)
Research encouraged by managers (n=177)	5 (2.8)	24 (13.6)	93 (52.5)	55 (31.1)
Research activity in the department (n=178)	5 (2.8)	15 (8.4)	91 (51.1)	67 (37.6)
Mentors available to guide (n=175)	8 (4.6)	20 (11.4)	80 (45.7)	67 (38.3)
Experienced internal colleagues able to supervise (n=175)	9 (5.1)	19 (10.9)	83 (47.4)	64 (36.6)
Experienced external colleagues able to supervise (n=175)	7 (4.0)	27 (15.4)	89 (50.9)	52 (29.7)
Opportunities to participate at your own level (n=175)	5 (2.9)	12 (6.9)	94 (53.7)	64 (36.6)
Desire to prove a theory or hunch (n=177)	3 (1.7)	17 (9.6)	106 (59.9)	51 (28.8)
Dedicated time for research (n=177)	10 (5.6)	13 (7.3)	59 (33.3)	95 (53.7)
Colleagues doing research (n=176)	8 (4.5)	14 (8.0)	100 (56.8)	54 (30.7)
Research written into role description (n=176)	11 (6.3)	22 (12.5)	82 (46.6)	61 (34.7)
Funding available (n=174)	10 (5.7)	19 (10.9)	61 (35.1)	84 (48.3)

References

1. College of Radiographers. *CoR Research Strategy 2021 – 26. Aiming to deliver research-based practice in radiography*; 2021. Available online: <https://www.collegeofradiographers.ac.uk/research-grants-and-funding/cor-research-strategy> [Accessed 25/10/2025].

2. European Federation for Radiography Societies (EFRS) Activity plan 2023–2025. Available online: <https://api.ehrs.eu/api/assets/posts/323> [Accessed 25/10/2025].

3. Oliveira M, Hogg P, Di Prospero L, Lacey S, El-Farra S, Johansen S. Research activity among diagnostic and therapeutic radiographers: an international survey. *J Med Imaging Radiat Sci*. 2024;55(2):232–243. <https://doi.org/10.1016/j.jmir.2024.02.005>.

4. Al Balushi H, Watts H, Akudjedu TN. Research and evidence-based practice in clinical radiography: a systematic review of barriers and recommendations for a new direction. *Radiography*. 2024;30(2):538–559. <https://doi.org/10.1016/j.radi.2024.01.012>.

5. Bolejko A, Musmann BR, Debess J, Henner A, Sanderud A, Andersson BT, Fridell K. Factors of importance for the development of evidence-based practice amongst radiographers in public healthcare. *Radiography (Lond)*. 2025;31(1):1–5. <https://doi.org/10.1016/j.radi.2024.10.011>.

6. Ooi C-C, Lee SH-E, Soh BP. A survey on the research awareness and readiness among radiographers in Singapore General Hospital (SGH). *Radiography*. 2012;18(4):264–269. <https://doi.org/10.1016/j.radi.2012.06.004>.

7. Paulo G. Radiographer research in radiation protection: national and European perspectives. *Radiography (Lond)*. 2020;26(Suppl 2):S42–S44. <https://doi.org/10.1016/j.radi.2020.05.006>.

8. Malamateniou C. Radiography and research: a United Kingdom perspective. *European Journal of Radiography*. 2009;1(1):2–6. <https://doi.org/10.1016/j.ejrad.2008.12.003>.

9. Jensen J, Andersen PB, Skjøth F, Pedersen MRV. An overview of radiography research and publication productivity in Denmark. *Radiography (Lond)*. 2025;31(2):102877. <https://doi.org/10.1016/j.radi.2025.01.012>.

10. Saukko E, Andersson BT, Bolejko A, Debess J, Fridell K, Henner A, Sanderud A. Radiographers' involvement in research activities and opinions on radiography research: a Nordic survey. *Radiography (Lond)*. 2021. <https://doi.org/10.1016/j.radi.2021.02.002>.

11. Vikestad KG, Hafskjold L, Kjellev E, Sebuødegård S, Hofvind S. Radiographers' opinions on radiography research in Norway – a national survey. *Radiography*. 2017;23(2):135–140. <https://doi.org/10.1016/j.radi.2016.12.006>.

12. Knight KJ, Beasley M, McConnell J, O'Regan T, Alexander CM, Donovan T, McNair HA. Research culture, barriers and facilitators within the radiography workforce in the UK – results of a national survey. *Radiography (Lond)*. 2025;31(4):102959. <https://doi.org/10.1016/j.radi.2025.102959>.

13. Yakubu A, Briggs E, Hacking S, Akudjedu TN. Clinical audit and research in radiography practice: an exploration of the English landscape. *Radiography (Lond)*. 2023 Jan;29(1):200–206. <https://doi.org/10.1016/j.radi.2022.11.004>. Epub 2022 Dec 5. PMID: 36481787.

14. Alakhras M, Al-Mousa DS, Al Mohammad B, Spuur KM. Knowledge, attitude, understanding and implementation of evidence-based practice among Jordanian radiographers. *Radiography*. 2023;29(4):760–766. <https://doi.org/10.1016/j.radi.2023.05.007>.

15. Bolejko A, Andersson BT, Debess J, Fridell K, Henner A, Sanderud A, Musmann BR. Facilitators for and barriers to radiography research in public healthcare in Nordic countries. *Radiography*. 2022;28(1):88–94. <https://doi.org/10.1016/j.radi.2021.08.007>.

16. Reid K, Edwards H. Evaluating the role of the diagnostic research radiographer. *Radiography*. 2011;17(3):207–211.

17. Dennett AM, Cauchi T, Harding KE, Kelly P, Ashby G, Taylor NF. Research interest, experience and confidence of allied health professionals working in medical imaging: a cross-sectional survey. *J Med Radiat Sci*. 2021;68(2):121–130. <https://doi.org/10.1002/jmrs.432>.

18. Ger N, Lugossy A-M, Petruca P. Status of “African research and its contribution to global health research: a review and an opinion”. *Discover Public Health*. 2024;21(1):192. <https://doi.org/10.1186/s12982-024-00323-6>.

19. Creswell JW, Plano Clark VL. *Designing and conducting mixed methods research*. 3rd Edition. Thousand Oaks, CA: Sage; 2017.

20. Hosmer DW, Lemeshow S, Sturdivant RX. *Applied logistic regression*. John Wiley & Sons; 2013.

21. Braun V, Clarke V. *Thematic analysis*. Los Angeles ; London ; New Delhi: Sage; 2022.

22. Fetters MD, Curry LA, Creswell JW. Achieving integration in mixed methods Designs—Principles and practices. *Health Serv Res*. 2013;48:2134–2156. <https://doi.org/10.1111/1475-6773.12117>.

23. Nightingale, J., Stewart-Lord, A. Mixed methodology: a pragmatic approach to investigating complexity in radiography research. *Radiography*, Volume 32, Issue 1, 103257.

24. Akudjedu TN, Reeve R, Eaton J, Costigan C, Burnett C, Barnsley H, McNair H, Simcock IC. Research and fellowship funding perspectives: a collective NIHR experience of a community of pre- and post-doctoral radiographers. 2025;31(2).

25. Snaith B, Harris M, Harris R. Radiographers as doctors: a profile of UK doctoral achievement. *Radiography*. 2016;22(4):282–286.

26. Al-Ghunaim T, Harrison G, Kaur E, Arezina J, Carr R, Johnson J. Facilitators and barriers to undertaking research into the practice and delivery of clinical ultrasound: a qualitative investigation. 2024;30(5):1391–1397.

27. Younas A, Fábregues S, Munce S, Creswell JW. Framework for types of meta inferences in mixed methods research. *BMC Med Res Methodol*. 2025 Jan 24;25(1):18. <https://doi.org/10.1186/s12874-025-02475-8>. PMID: 39856568; PMCID: PMC11758751.

28. Peng Y, Xiao SW, Tu H, Xiong XY, Ma ZJ, Xu WJ, Cheng T. The Impostor phenomenon Among nursing students and nurses: a scoping review. *Front Psychol*. 2022 Mar 9;13:809031. <https://doi.org/10.3389/fpsyg.2022.809031>. PMID: 35356345; PMCID: PMC8959846.